Convergence Tests

The ratio and comparison tests

1 Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

2 Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

3 Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.

Alternating series and the p-series test

4 Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

5 Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.

The integral test

6 Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x}\, dx$.

7 Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2}\, dx$.

Choosing the right test

8 For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum \frac{2^n}{n!}$

b $\sum \frac{(-1)^n}{\sqrt{n}}$

c $\sum \frac{1}{n^4}$

The limit comparison test

9 Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.

10 Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2 - 1}{n^4 + 2n}$, comparing to $\sum \frac{1}{n^2}$.